# Experimenting with zero-models

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Slides: https://www.marialoni.org/resources/3BNU26.pdf

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# Two views

- People draw conclusions beyond what is literally said
- The standard view: enriched interpretations (aka implicatures) result from an alternative-based mechanism (slow, deliberative, costly)
  - (i) computation of literal meaning; (ii) recognition of alternative phrases which could have been used; (iii) negation of these alternatives; ...
- The Nihil hypothesis: enrichments sometimes result from biases due to our cognitive limitations, notably: (fast, not deliberative, effortless)
  1. **neglect-zero**: a tendency to avoid emptiness
  2. **no-split**: a negative bias towards the parallel processing of alternatives

## Today

- Experimenting with zero-models: 3 recent studies testing the predictions of these two views

## Focus on three inferences

(1) a. Some of the squares are red ⇒ not all of the squares are red [scalar **UB**]
b. Each square is white or red ⇒ there are white squares and red squares [**DIST**]
c. Less than 3 squares are white ⇒ there are some white squares [**ESQ**]

| | **UB** | **DIST** | **ESQ** |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based[1] | alt-based [?] |
| NIHIL | — | neglect-zero | neglect-zero |

[1] Crnič, Chemla, & Fox (2015); Bar-Lev & Fox (2023).

# The Nihil hypothesis

NEGLECT-ZERO
When interpreting a sentence speakers construct models depicting reality and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*)

## Illustrations

(2) Less than three squares are black. **[ESQ]**

a. Verifier: [■, □, ■] $\rightsquigarrow_{nz}$ there are black squares
b. Falsifier: [■, ■, ■]
c. Zero-models: [□, □, □]; [■, ■, ■]; . . .

Zero-models in (2-c) verify the sentence by virtue of an empty set of black squares

(3) Each square is white or red. **[DIST]**

a. Verifier: [■, □, ■] $\rightsquigarrow_{nz}$ there are white squares & red squares
b. Falsifier: [■, □, ■]
c. Zero-models: [□, □, □]; [■, ■, ■]

Zero-models in (3-c) verify the sentence by virtue of an empty set of white sqrs or of red sqrs

NZ hypothesis: neglect-zero at the origin of many common departures from classical reasoning

- Free choice and ignorance [MA 2022]
- Existential Import, Aristotle's Thesis and Boethius' Thesis
- Inferences from quantifiers: **ESQ, DIST** [TODAY]

# qBSML: adding quantifiers to BSML

- A quantified modal language interpreted in standard Kripke models
- Formulas interpreted wrt sets of points of evaluation (*teams*)
- Teams defined as **sets of world-assignment pairs**

**Language:**
$t ::= c|v$
$\phi ::= P^n(\vec{t}) \mid \neg\phi \mid \phi \vee \psi \mid \phi \wedge \psi \mid \exists v\phi \mid \forall v\phi \mid \Box\phi \mid \text{NE}$

**Model:**
$\mathcal{M} = \langle W, D, R, I \rangle$

**Team/Information State:**
A state is set of indices $i = \langle w_i, g_i \rangle$, where $w_i \in W$ and $g_i$ is a variable assignment function

# qBSML: Quantified Modal Logic + NE

$$
\begin{array}{lll}
\mathcal{M}, s \models P^n t_1 \dots t_n & \text{iff} & \text{for all } i \in s : \langle [t_1]_{\mathcal{M},i}, \dots, [t_n]_{\mathcal{M},i} \rangle \in I(w_i)(P^n) \\
\mathcal{M}, s =\!| P^n t_1 \dots t_n & \text{iff} & \text{for all } i \in s : \langle [t_1]_{\mathcal{M},i}, \dots, [t_n]_{\mathcal{M},i} \rangle \notin I(w_i)(P^n) \\
\mathcal{M}, s \models \neg\varphi & \text{iff} & \mathcal{M}, s =\!| \varphi \\
\mathcal{M}, s =\!| \neg\varphi & \text{iff} & \mathcal{M}, s \models \varphi \\
\mathcal{M}, s \models \varphi \vee \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \text{ and } \mathcal{M}, t \models \varphi \text{ and } \mathcal{M}, t' \models \psi \\
\mathcal{M}, s =\!| \varphi \vee \psi & \text{iff} & \mathcal{M}, s =\!| \varphi \text{ and } \mathcal{M}, s =\!| \psi \\
\mathcal{M}, s \models \varphi \wedge \psi & \text{iff} & \mathcal{M}, s \models \varphi \text{ and } \mathcal{M}, s \models \psi \\
\mathcal{M}, s =\!| \varphi \wedge \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \text{ and } \mathcal{M}, t =\!| \varphi \text{ and } \mathcal{M}, t' =\!| \psi \\
\mathcal{M}, s \models \Box\varphi & \text{iff} & \text{for all } i \in s : \mathcal{M}, R(w_i)[g_i] \models \varphi \\
\mathcal{M}, s =\!| \Box\varphi & \text{iff} & \text{for all } i \in s : \exists X \subseteq R(w_i) \text{ and } X \neq \varnothing \text{ and } \mathcal{M}, X[g_i] =\!| \varphi \\
\mathcal{M}, s \models \mathtt{NE} & \text{iff} & s \neq \varnothing \qquad [X[g_i] = \{\langle w, g_i \rangle \mid w \in X\}] \\
\mathcal{M}, s =\!| \mathtt{NE} & \text{iff} & s = \varnothing \qquad [R(w_i) = \{v \in W \mid w_i R v\}] \\
\mathcal{M}, s \models \forall x\varphi & \text{iff} & \mathcal{M}, s[x] \models \varphi \\
\mathcal{M}, s =\!| \forall x\varphi & \text{iff} & \mathcal{M}, s[x/h] =\!| \varphi, \text{ for some } h : s \to \wp(D) \setminus \varnothing \\
\mathcal{M}, s \models \exists x\varphi & \text{iff} & \mathcal{M}, s[x/h] \models \varphi, \text{ for some } h : s \to \wp(D) \setminus \varnothing \\
\mathcal{M}, s =\!| \exists x\varphi & \text{iff} & \mathcal{M}, s[x] =\!| \varphi
\end{array}
$$

# qBSML: the universal quantifier

- A state $s$ supports $\forall x\phi$ iff $s[x]$, the universal $x$-extension of $s$, supports $\phi$:

$$\mathcal{M}, s \models \forall x\varphi \text{ iff } \mathcal{M}, s[x] \models \varphi$$

- The universal $x$-extension of $s \mapsto s$ updated with all possible values for $x$

$$s = \boxed{\langle w, \lambda \rangle} \mapsto s[x] = \begin{array}{|c|}\hline \langle w, x \to a \rangle \\ \hline \langle w, x \to b \rangle \\ \hline \langle w, x \to c \rangle \\ \hline \langle w, x \to d \rangle \\ \hline \dots \\ \hline \end{array}$$

Figure: An initial max info state and its universal $x$-extension, $D = \{a, b, c, d, \dots\}$

# Neglect-zero effects in qBSML: neglect-zero enrichment

## Neglect-zero enrichment

For NE-free $\alpha$, $[\alpha]^+$ defined as follows:

$$
\begin{aligned}
[Pt_1 \dots t_n]^+ &= Pt_1 \dots t_n \wedge \mathrm{NE} \\
[\neg\alpha]^+ &= \neg[\alpha]^+ \wedge \mathrm{NE} \\
[\alpha \vee \beta]^+ &= ([\alpha]^+ \vee [\beta]^+) \wedge \mathrm{NE} \\
[\alpha \wedge \beta]^+ &= ([\alpha]^+ \wedge [\beta]^+) \wedge \mathrm{NE} \\
[\Box\alpha]^+ &= \Box[\alpha]^+ \wedge \mathrm{NE} \\
[\forall x\alpha]^+ &= \forall x[\alpha]^+ \wedge \mathrm{NE} \\
[\exists x\alpha]^+ &= \exists x[\alpha]^+ \wedge \mathrm{NE}
\end{aligned}
$$

$[\ ]^+$ enriches formulas with the requirement to satisfy NE distributed along each of their subformulas

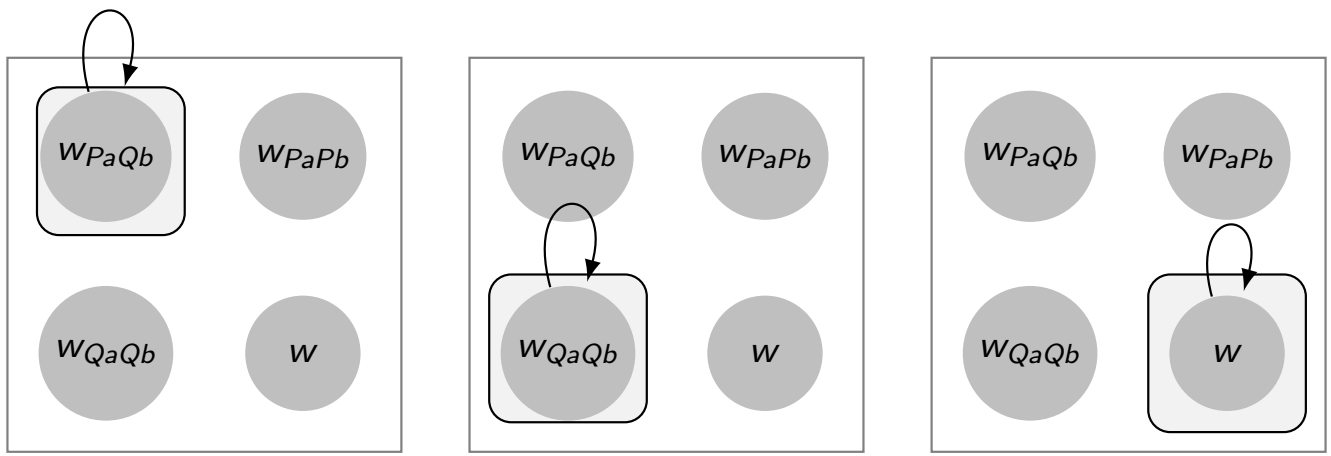
# qBSML: illustration

## Interaction disjunction and quantifiers: distributive inference

$$D = \{a, b\}$$

$$s = \boxed{\langle w_{PaQb}, \lambda \rangle} \models [\forall x(Px \vee Qx)]^+ \text{ since } s[x] = \boxed{\begin{array}{c}\langle w_{PaQb}, x \to a \rangle \\ \langle w_{PaQb}, x \to b \rangle\end{array}} \models [Px \vee Qx]^+$$

$$s = \boxed{\langle w_{QaQb}, \lambda \rangle} \not\models [\forall x(Px \vee Qx)]^+ \text{ since } s[x] = \boxed{\begin{array}{c}\langle w_{QaQb}, x \to a \rangle \\ \langle w_{QaQb}, x \to b \rangle\end{array}} \not\models [Px \vee Qx]^+$$




No-zero verifier
$\models \forall x(Px \vee Qx)$
$\models [\forall x(Px \vee Qx)]^+$

**Zero-model**
$\models \forall x(Px \vee Qx)$
$\not\models [\forall x(Px \vee Qx)]^+$

Falsifier
$=\!|\ \forall x(Px \vee Qx)$
$=\!|\ [\forall x(Px \vee Qx)]^+$

Figure: Models for $\forall x(Px \vee Qx)$

(4) Everything is black or white $\rightsquigarrow_{nz}$ there are black things and white things
$[\forall x(Px \vee Qx)]^+ \models \exists x Px \wedge \exists x Qx$ (with max info)

# Neglect-zero effects on quantifiers

## Predictions of qBSML$^+$ (with max info & $\phi \to \psi = \neg\phi \vee (\phi \wedge \psi)$)

(5) Less than three squares are black $\mapsto \forall xyz((Sx \wedge Bx \wedge \dots) \to (x = y \vee \dots))$

a. Verifier: [■, □, ■]
b. Falsifier: [■, ■, ■]
c. Zero-models: [□, □, □]; [▲, ▲, ▲]; ... $\rightsquigarrow_{nz}$ there are black squares (ESQ)

(6) Every square is black. $\mapsto \forall x(Sx \to Bx)$

a. Verifier: [■, ■, ■]
b. Falsifier: [■, □, ■]
c. Zero-models: [△, △, △]; [▲, ▲, ▲]; ... $\rightsquigarrow_{nz}$ there are squares (ESQ-restr)

(7) No squares are black. $\mapsto$ (i) $\forall x(Sx \to \neg Bx)$; (ii) $\neg\exists x(Sx \wedge Bx)$

a. Verifier: [□, □, □]
b. Falsifier: [■, □, □]
c. Zero-models for (i): [△, △, △]; [▲, ▲, ▲]; ... $\rightsquigarrow_{nz}$ there are squares
d. Zero-models for (ii): none no neglect-zero effect

(8) Each square is white or red. $\mapsto \forall x(Sx \to (Wx \vee Rx))$

a. Verifier: [■, □, ■]
b. Falsifier: [■, □, ■]
c. Zero-models: [■, ■, ■]; [□, □, □] $\rightsquigarrow_{nz}$ there are white sqrs & red sqrs

These predictions tested in 3 recent experiments

# Experimenting with zero-models: the team



Oliver Bott



Fabian Schlotterbeck



Sonia Ramotowska



Tomasz Klochowicz

# Experimenting with zero-models

## Three inferences

(9) a. Some of the squares are red ⇒ not all of the squares are red [UB]
b. Fewer than 3 squares are white ⇒ there are some white squares [ESQ]
c. Each square is white or red ⇒ there are white squares and red squares [DIST]

## Two competing accounts

| | UB | ESQ | DIST |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based [?] | alt-based |
| NIHIL | — | neglect-zero | neglect-zero |

How to test these different hypotheses?

## Strategy A

- Much is known about UB scalar implicatures:
  - Delay effect: $RT_{enriched} > RT_{literal}$ [L. Bott & Noveck, 2004]
  - Lower implicature-rate under cognitive load [De Neys & Schaeken, 2007]
  - Acquired late [Noveck, 2001]
  - Lower rate in questions
  - ...
- Do these features generalise to other inferences in (9)?
  - Standard view: in principle YES (all derive from single alt-based mechanism)
  - NIHIL: NO (nz effects result from a preference to minimize cognitive effort)
- Strategy A: valuable and instructive but not resolutive on whether one and the same mechanism derives all inferences in (9)

# Experimenting with zero-models

## Three inferences

(10) a. Some of the squares are red ⇒ not all of the squares are red [UB]
b. Fewer than 3 squares are white ⇒ there are some white squares [ESQ]
c. Each square is white or red ⇒ there are white squares and red squares [DIST]

## Two competing accounts

| | UB | ESQ | DIST |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based [?] | alt-based |
| NIHIL | — | neglect-zero | neglect-zero |

How to test these different hypotheses?

## Strategy B

- **Priming:** two assumptions (L. Bott and Chemla 2016)
  - Rates of enrichments (implicature-based responses) can systematically be affected by manipulations in a priming phase
  - If one single cognitive mechanism is responsible for deriving two different inferences, drawing/suspending one on a prime trial should increase/decrease the rate of the other in the subsequent trial
- Will inference-rate change after participants are primed to suspend other inferences in (10)?
  - Standard view: YES overall
  - NIHIL: only between ESQ & DIST

# Experimenting with zero-models

## Three inferences

(11) a. Some of the squares are red ⇒ not all of the squares are red [UB]
b. Fewer than 3 squares are white ⇒ there are some white squares [ESQ]
c. Each square is white or red ⇒ there are white squares and red squares [DIST]

## Two competing accounts

| | UB | ESQ | DIST |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based [?] | alt-based |
| NIHIL | — | neglect-zero | neglect-zero |

## Three experiments

- Exp 1: Answering questions about the empty set (O. Bott *et al*, SuB24) [StrA]
- Exp 2: Priming with zero-models (Klochowicz *et al*, CogSci 2025, SuB25) [StrB]
- Exp 3: Inferences under cognitive load (Ramotowska & MA, XPRAG25) [StrA]

## Three main conclusions

(a) Evidence that DIST differs from UB (Exp 1, Exp 2, Exp 3)

(b) Mixed results concerning ESQ (Exp 3) ⇐

(c) Evidence that DIST & ESQ (but not UB) involve the same cognitive process (Exp 2)

**Tentative conclusion** While (a) in principle explainable by both views (via lexical access) and (b) by neither view, (c) is a challenge for the standard implicature view

# Experimenting with zero-models

## Three inferences

(12)
a. Some of the squares are red ⇒ not all of the squares are red [UB]
b. Fewer than 3 squares are white ⇒ there are some white squares [ESQ]
c. Each square is white or red ⇒ there are white squares and red squares [DIST]

## Two competing accounts

| | UB | ESQ | DIST |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based [?] | alt-based |
| NIHIL | — | neglect-zero | neglect-zero |

## Task

All experiments involved a picture verification task and compared:

- Target: literal reading ↦ **true** & enriched reading ↦ **false**

[■, ■, ■, ■, ■]

- True-control: literal reading ↦ **true** & enriched reading ↦ **true**

[■, ■, ■, □, □]

- False-control: literal reading ↦ **false** & enriched reading ↦ **false**

[■, ■, □, □, □]

# Exp1: Answering questions about the empty set (Bott et al, SuB 2024)

## Four inferences

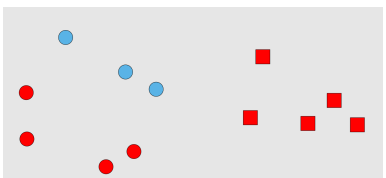
(13)
a. Some of the squares are black ⇒ not all of the squares are black [UB]
b. Each square is red or white ⇒ there are white squares and red squares [DIST]
c. Less than 3 squares are black ⇒ there are some black squares [ESQ-**scope**]
d. Less than 3/every/no squares are black ⇒ there are some squares [ESQ-**restrictor**]

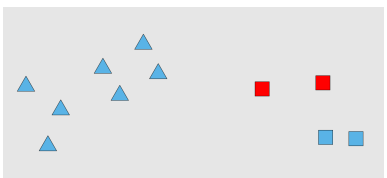
## Three competing accounts

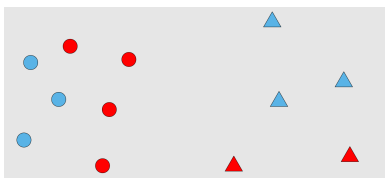
| | UB | DIST | ESQ-scope | ESQ-restrictor |
|---|---|---|---|---|
| Standard implicature view | alt-based | alt-based | alt-based [?] | — |
| O. Bott *et al* 2019 | — | — | neglect-zero | presupposition |
| NIHIL | — | neglect-zero | neglect-zero | neglect-zero |

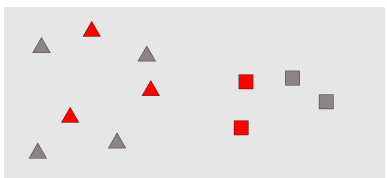
- Question-answer task (German)

(14) Ist jedes Dreieck entweder rot oder blau? Ja/Nein/Komische Frage
(Is every triangle either red or blue?) Yes/No/Odd question


empty restr


DIST target (zero-model)


control '*yes*'


control '*no*'

- Main results:
  1. Evidence that ESQ-restrictor is a presupposition: questions in empty restrictor models uniformly (more than 90%) perceived as odd

↦ challenge for NIHIL

# Exp1: Answering questions about the empty set (Bott et al, SuB 2024)

## Four inferences

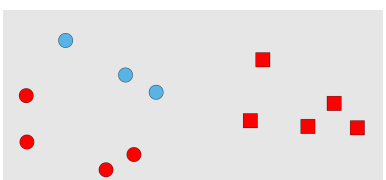
(15)
a. Some of the squares are black ⇒ not all of the squares are black [UB]
b. Each square is red or white ⇒ there are white squares and red squares [DIST]
c. Less than 3 squares are black ⇒ there are some black squares [ESQ-**scope**]
d. Less than 3/every/no squares are black ⇒ there are some squares [ESQ-**restrictor**]

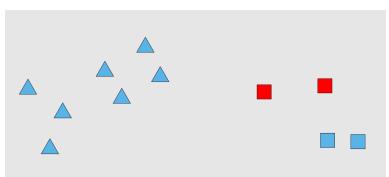
## Three competing accounts

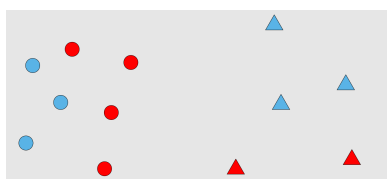
| | UB | DIST | ESQ-scope | ESQ-restrictor |
|---|---|---|---|---|
| Standard implicature view | alt-based | alt-based | alt-based [?] | — |
| O. Bott *et al* 2019 | — | — | neglect-zero | presupposition |
| NIHIL | — | neglect-zero | neglect-zero | neglect-zero |

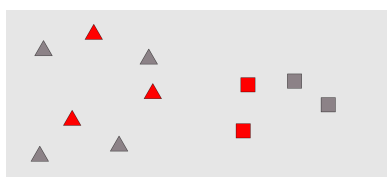
- Question-answer task:

(16) Ist jedes Dreieck entweder rot oder blau? Ja/Nein/Komische Frage
(Is every triangle either red or blue?) Yes/No/Odd question


empty restr


DIST target (zero-model)


control '*yes*'


control '*no*'

- Main results:
  2. Using results from previous studies: ESQ-scope (37%) & DIST (23%) unaffected by question environment; UB much less available (10%, while 40% when unembedded)
  3. Inconclusive evidence on whether ESQ-scope & DIST have the same source

# Exp2: Priming with zero-models (Klochowicz *et al*, CogSci 2025, SuB25)

## Four inferences

(17)
a. Some of the squares are black ⇒ not all of the squares are black [UB]
b. Each square is red or white ⇒ there are white and red squares [DIST]
c. At most 2 squares are black ⇒ there are some black squares [**ESQ-s**]
d. Fewer than 3 squares are black ⇒ there are some black squares [ESQ]

## Two competing accounts

| | UB | DIST | ESQ(-s) |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based | alt-based [?] |
| NIHIL | — | neglect-zero | neglect-zero |

## The priming study

- 3 sub-experiments tested whether frequency of ESQ changed after participants were primed to suspend other enrichments in (17-a)-(17-c)
- Predictions/expectations:

| | Implicature view | NIHIL |
|---|---|---|
| DIST ⇒ ESQ | yes [?] | yes |
| UB ⇒ ESQ | yes [?] | no |
| ESQ-s ⇒ ESQ | yes [?] | yes |

# Exp2: Better-picture paradigm (adapted from L. Bott & Chemla 2016)

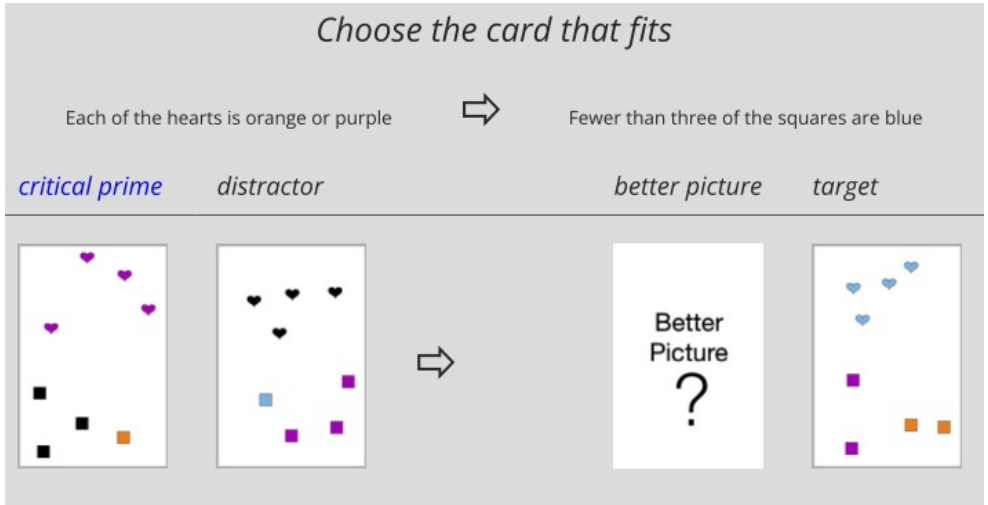



Figure: DIST ⇒ ESQ

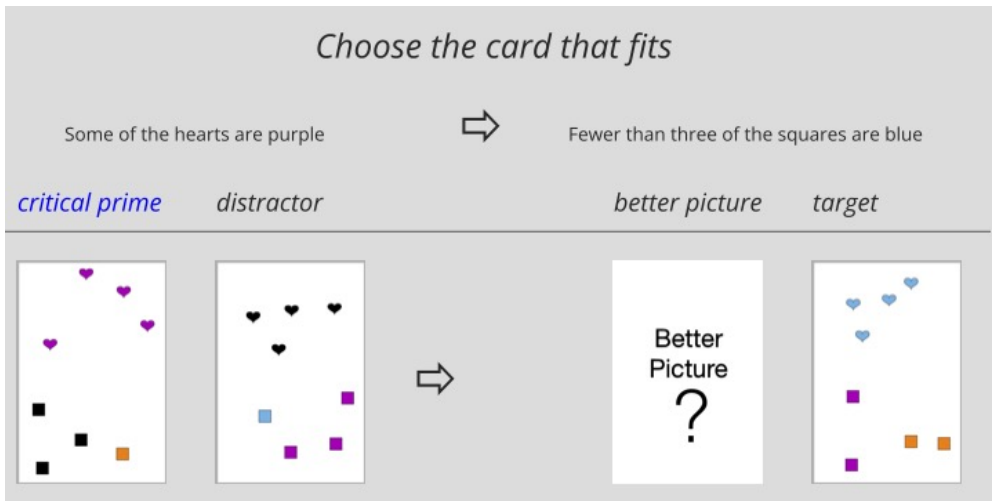
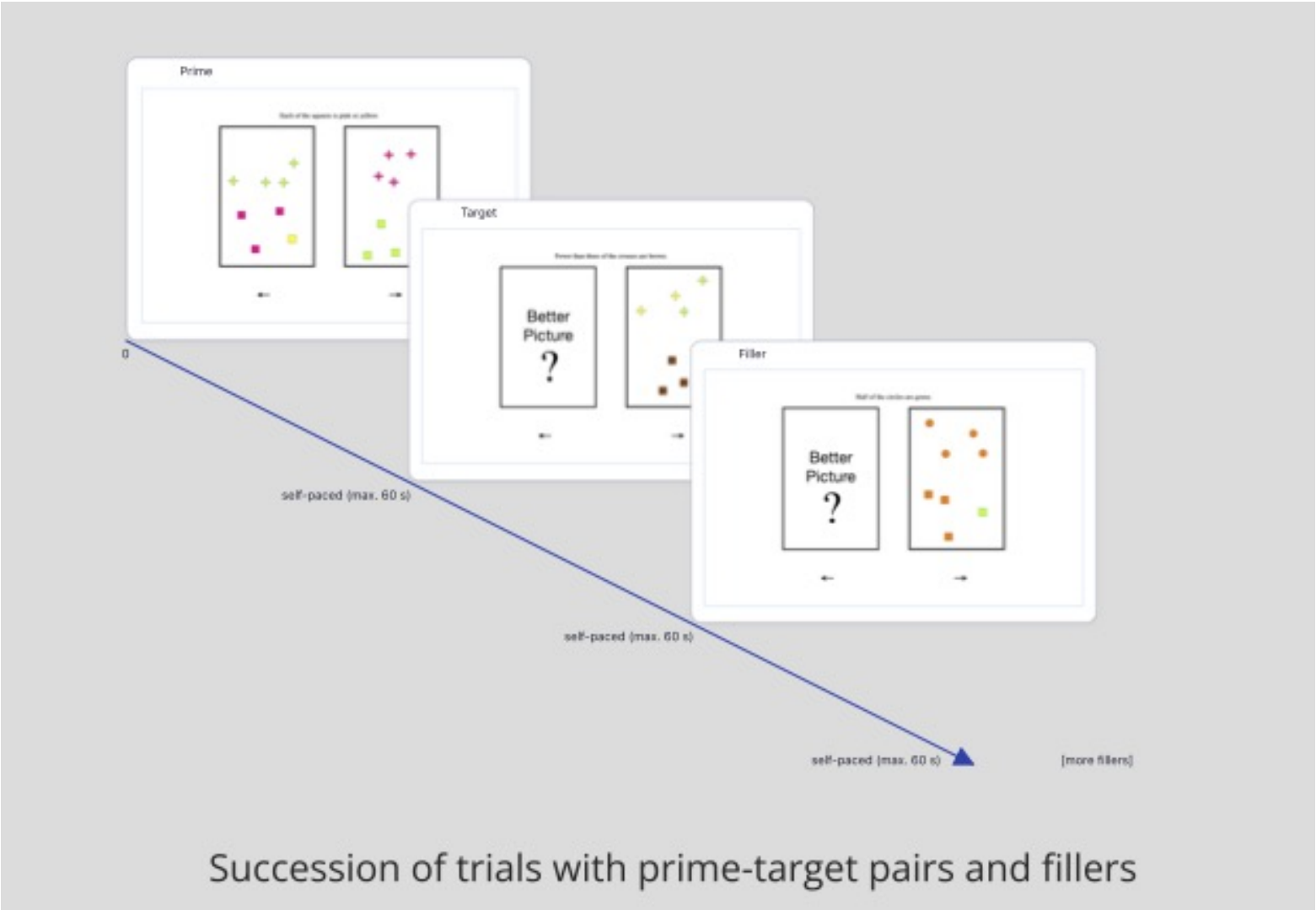


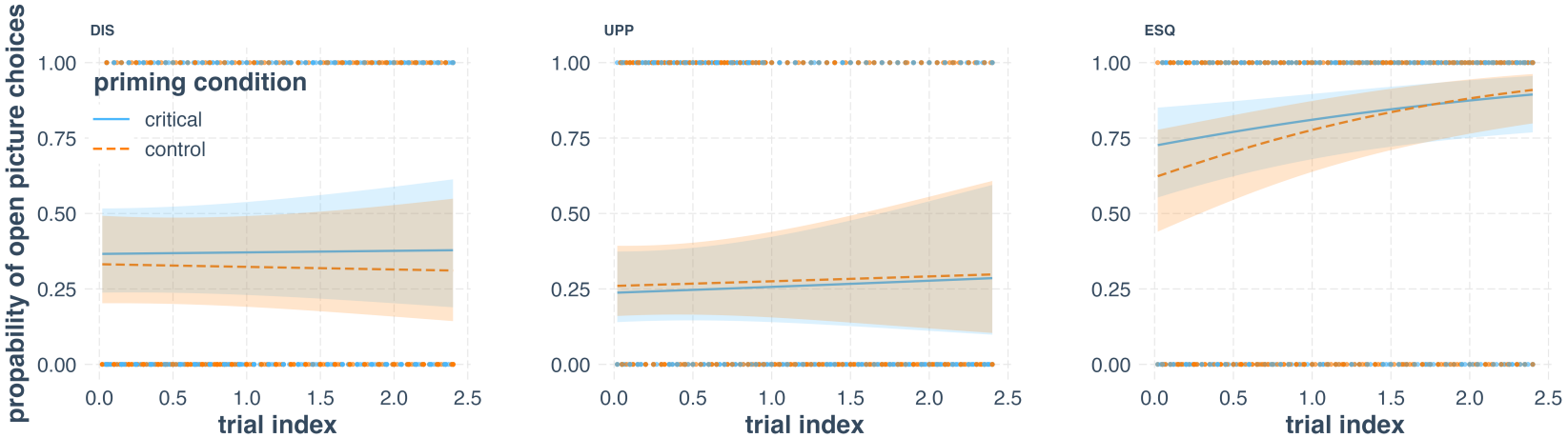

Figure: UB ⇒ ESQ

# Exp2: Procedure




Succession of trials with prime-target pairs and fillers

# Exp2: Summary of main findings




## Results

1. DIST ⇒ ESQ: trial to trial priming
2. UB ⇒ ESQ: no priming and overall lower rate of zero-model choices
3. ESQ-s ⇒ ESQ: no trial-to-trial priming but global adaptation (spill-over) at ceiling towards end of experiment

## Discussion

- Evidence of a common mechanism in use for deriving DIST & ESQ inferences (sub-exp 1), but not UB scalar implicatures (sub-exp 2)
  ↦ a challenge for standard view
- But results in sub-exp 3 puzzling (follow-ups planned)

# Exp3: Inferences under cognitive load (Ramotowska & MA, XPRAG25)

## Four inferences

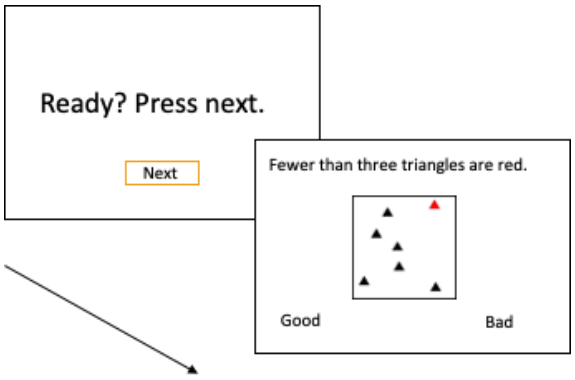
(18)
a. Some of the squares are black ⇒ not all of the squares are black [UB]
b. Fewer than 3 squares are black ⇒ there are some black squares [ESQ]
c. Each square is white or red ⇒ there are white squares and red squares [DIST]
d. White squares are scarce ⇒ white squares are not absent **[LB]**

## Two competing accounts

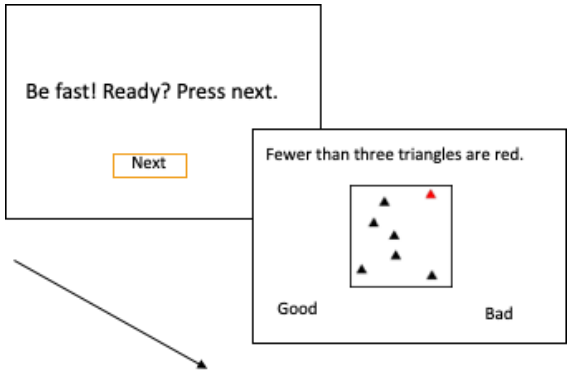
| | UB | ESQ | DIST | LB |
|---|---|---|---|---|
| Standard implicature view | alt-based | alt-based [?] | alt-based | alt-based |
| NIHIL | — | neglect-zero | neglect-zero | neglect-zero [?] |

- Background: different signatures of processing cost of implicatures:
  - Reaction Time (RT): delay (D) vs reverse delay (R) effect
  - Change in inference-rate under cognitive load (speed-pressure/dual task)
- The study: we measured RT & inference-rate in 3 sub-exps with same verification task, but different manipulations:
  - (i) baseline (BAS); (ii) speed-pressure (SP); (iii) dual task (WM).
- Predictions/expectations:
  - Alt-based implicatures: more costly than literal responses
    RT: Delay effect & increase (↑) of literal responses under load
  - Neglect-zero effects: less costly than literal responses
    RT: Reverse delay effect & decrease (↓) of literal responses under load

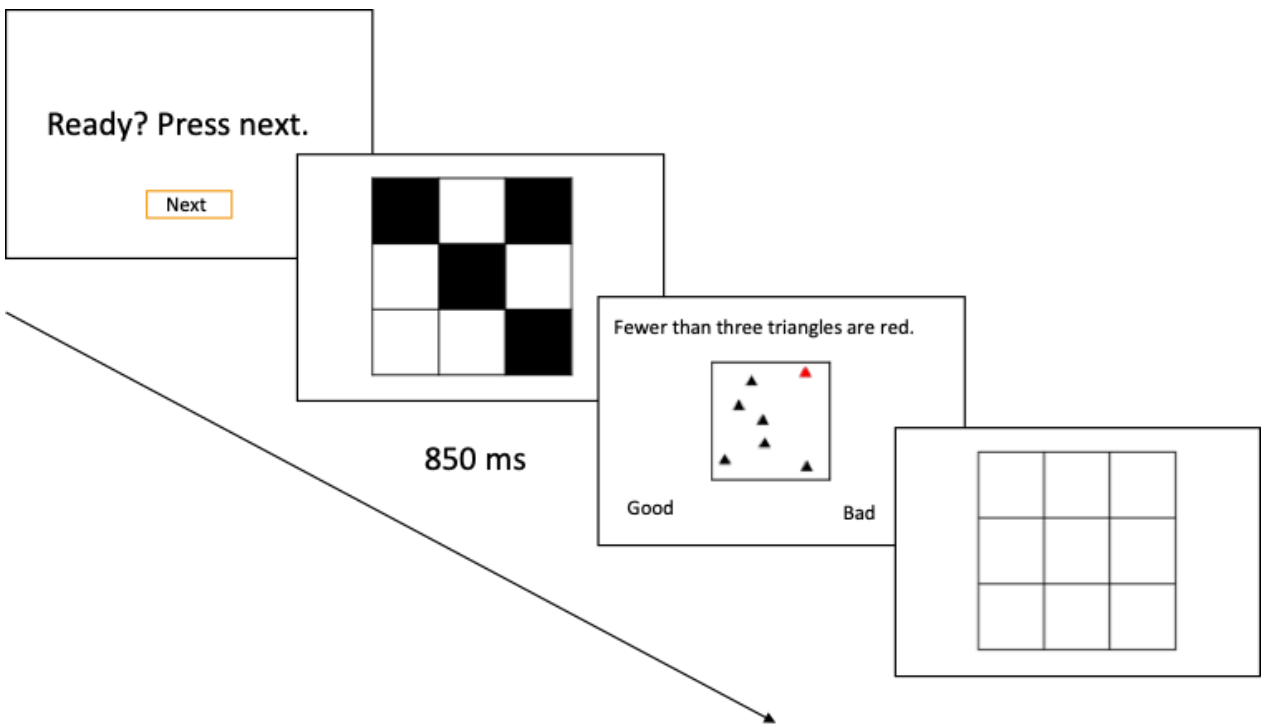
# Exp3: Same verification task, different manipulations




(a) Baseline (no load)




(b) Speed-stressing instruct. (SP)




(c) Dual-task (WM)

# Results exp3: target acceptance under cognitive load

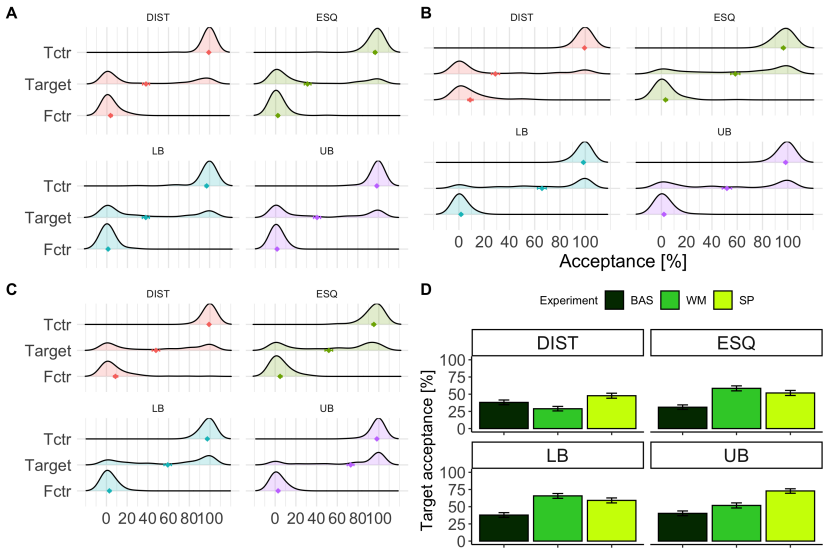



Figure: Acceptance rate in **A** BASELINE, **B** WM, **C** SP by conditions: Tctr (*true* control), Target, Fctr (*false* control). **D** target acceptance (literal responses) in each sub-exp

More literal responses under cognitive load for all cases except **DIST**:

- Fewer **UB** under cognitive load (replication)
- Similar results for **LB** ($\neq$ vTiel et al 2019) & **ESQ** (new)
- Different patterns for **DIST**: inference-rate unaffected by manipulation

ESQ result $\mapsto$ a challenge for NIHIL

# Results exp3: Reaction Time

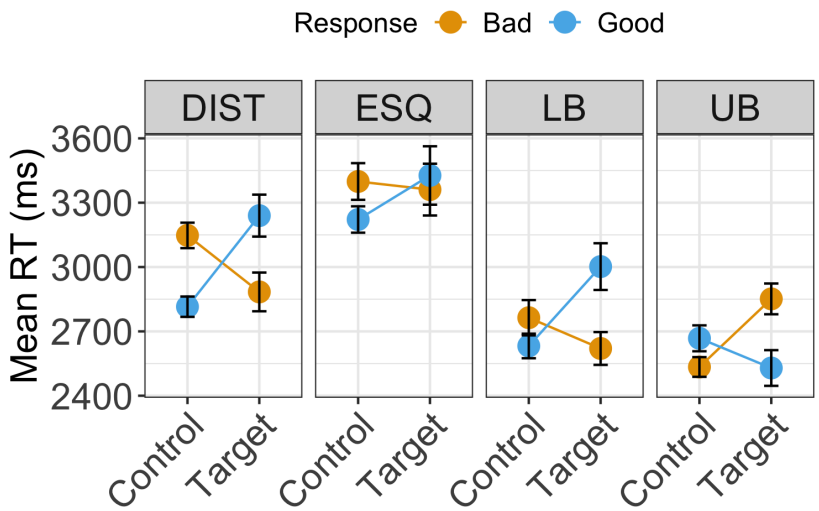



Figure: Mean reaction times (in ms) in BAS sub-experiment.

- Replicated classical Delay effect for **UB**:
  - target false > target true **enriched (false) slower than literal**
  - target false > false control
- Different patterns for **LB**, **ESQ** & **DIST**: Reverse delay effect
  - target true > target false **literal (true) slower than enriched**
  - target true > true control

# Exp3: Summary of main findings

| | UB | ESQ | LB | DIST |
|---|---|---|---|---|
| BAS | D | R | R | R |
| SP | $\uparrow$ D | $\uparrow$ R | $\uparrow$ N | $=$ R |
| WM high | $\uparrow$ | $\uparrow$ | $\uparrow$ | $=$ |
| WM low | $=$ | $=$ | $=$ | $=$ |

Table: Abbreviations: Reaction times: D - delay effect; R - Reverse delay effect; N - neither; * - marginal effect. Effects of manipulation: $\uparrow$ - higher rate of literal response; $\downarrow$ - lower rate of literal response; $=$ no effect.

## Results

- Reaction Time:
  **UB**: delay effect; **ESQ, LB & DIST**: reverse delay effect
- Cognitive load:
  **UB, ESQ & LB**: more literal responses ($\uparrow$); **DIST**: no effect ($=$)

## Discussion

- Different results for different signatures of cognitive cost:
  - RT results in agreement with Nihil hypothesis: UB | ESQ LB DIST
  - Manipulation results closer to implicature view: UB ESQ LB | DIST
- Challenges for both approaches:
  - Nihil: cognitive load results
  - Standard view: reverse delay effects; alt-based generation of ESQ

# Experimenting with zero-models: general discussion

## Inferences and competing hypothesis

(19)
a. Some of the squares are black ⇒ not all of the squares are black [UB]
b. Fewer than 3 squares are black ⇒ there are some black squares [ESQ]
c. Each square is white or red ⇒ there are white squares and red squares [DIST]

| | UB | ESQ | DIST |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based [?] | alt-based |
| NIHIL | — | neglect-zero | neglect-zero |

## Contrasting results from different experiments

1. Inference-rate in questions (exp1) UB | ESQ DIST
2. Priming (exp2) UB | ESQ DIST
3. RT results (exp3) UB | ESQ DIST
4. Cognitive load (exp3) UB ESQ | DIST
5. Acquisition (**new exp**): UB (late) vs ESQ (early) UB | ESQ

## Challenges for both views

- Standard implicature view: explains 4 (assuming lexical access), but leaves the rest unexplained & alt-based derivation of ESQ is unclear
- NIHIL: explains 1-3, & 5 but leaves the cognitive load data unexplained

**Main challenge for Nihil**: explain increase of literal ($0 < 3$) responses for ESQ under Speed Pressure/Dual-Task

# Cognitive load (dual-task) vs reaction time (RT)

## RT results do not align with cognitive load findings

- RT results: $\mapsto$ explainable by neglect-zero
  - UB: delay effect (slower than literal interpretation)
  - LB, ESQ, DIST: reverse delay effect (faster than literal interpretation)
- Cognitive load results: $\mapsto$ challenging for neglect-zero
  - UB, LB, ESQ: higher rate of literal interpretations under load
  - DIST: equal rate of literal interpretations under load

- Marty & Chemla 2013, Marty *et al*, 2020; Marty *et al*, 2024:
  Dual-task and response time results need not pattern together and, consequently, the two measures may reflect distinct cognitive effects
  - RT may reveal which interpretations are derived faster than others
  - Dual-task studies may reveal what resources are involved in the derivation of interpretations, independent of when these interpretations are derived and accessed

## On the cost of zero-models: a speculation

Neglect-zero interpretations accessed first, but literal (zero-model) responses do not necessarily come with additional demands on cognitive (WM) resources (do not involve any reasoning or access to lexical alternatives)

| processing stage | **scalar implicatures** | **neglect-zero effects** |
|---|---|---|
| stage 1 | literal | enriched |
| stage 2 | enriched (+WM cost) | literal |

# RT findings: reverse delay explained by neglect-zero

(20)
a. Some of the squares are red ⇒ not all [UB]
b. Fewer than 3 squares are white ⇒ some [ESQ]
c. Each square is white or red ⇒ some [DIST]
d. White squares are scarce ⇒ some [LB]

(21) **Target**: [■, ■, ■, ■, ■] [target false ↦ **enriched** & target true ↦ **literal**]

- UB: delay effect **[enriched > literal]**
  1. target false > target true
  2. target false > control false

  Standard explanation: extra cost of implicature calculation, *contra* defaultism
- LB, ESQ, DIST: reverse delay effect **[literal > enriched]**
  1. target true > target false
  2. target true > control true
- Neglect-zero explanation of reverse delay effects:
  - In target, ESQ, DIST & LB sentences judged true by virtue of an empty set ↦ require engaging and integration of a zero-model
- Nihil predictions:
  1. target true (zero-model used) > target false (zero-model not used)
  2. target true (zero-model used) > control true (no-zero verifier)
- Standard implicature view: reverse delay effects more challenging
  - Scalarity hypothesis (van Tiel *et al* 2019): SIs are cognitively demanding insofar as they introduce **negative** information ↦ only UB
- Possible explanation of reverse delay using generalisation of scalarity:
  1. target true > target false (positive SI) ???
  2. target true > control true ???

  ⇑ **extra cost literal reading unexplained without invoking defaultism!**

# Conclusion

- Three inferences & two views

| | UB | DIST | ESQ |
|---|---|---|---|
| Standard implicature view | alt-based | alt-based | alt-based[?] |
| NIHIL | — | neglect-zero | neglect-zero |

- Modelling zero-models in team semantics: $\text{BSML}^+$, $\text{qBSML}^+$

  Literal meanings (classical) + neglect-zero (NE) $\Rightarrow$ ESQ, DIST, FC, possibility, Aristotelian principles, LB, . . . , **but not UB**

- Experimenting with zero-models: 3 experimental studies
  - Questions, priming & RT findings: in agreement with NIHIL hypothesis
  - Cognitive load findings: challenging for both views, but mostly for NIHIL

- New testable predictions arising from speculative discussion. E.g. under cognitive load we predict:
  - less increase of literal interpretations for more complex non-numerical ESQs;
  - more homogeneous interpretations (no-zero & no-split).

THANK YOU![2]

[2]This work is supported by NWO OC project *Nothing is Logical* (grant no 406.21.CTW.023).